

Victor D. Masters

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Professional Summary

Field Robotics & Automation Specialist with 15+ years of experience engineering ruggedized SCADA (Supervisory Control and Data Acquisition) systems, embedded IoT (Internet of Things) platforms, and full-stack automation solutions for outdoor commercial jobsites. Architected custom data pipelines and control systems that quadrupled oil & gas production, driving a \$23M revenue increase and reducing operational downtime by 75% through real-time infrastructure monitoring and inspection automation. Expert in bridging hardware and software domains to deliver turnkey autonomous solutions, leveraging deep proficiency in ESP32 firmware, AI/ML (Artificial Intelligence/Machine Learning) inference, and heavy equipment operations to accelerate Industry 4.0 deployment in dynamic field environments.

Professional Experience

Ohio Valley Energy Systems | Austintown, OH Field Operations & Automation Lead | January 2008 – Present *

Architected and deployed two full-stack SCADA/telemetry platforms using embedded C, Python, and ESP32/ARM processors, quadrupling production output and generating a \$23M revenue increase over two years through optimized infrastructure monitoring and data-driven site selection. * Spearheaded standardization of maintenance protocols and developed benchmark testing procedures for field crews, reducing production downtime from 30–45 days to under 10 days annually while sustaining a 100% safety record across drilling, production, and restoration operations. * Designed and installed ruggedized automation hardware, pneumatic controls, and sensor networks for outdoor commercial jobsites, integrating real-time data pipelines and remote monitoring capabilities to enable inspection automation and predictive asset management. * Engineered data-mining algorithms and 3D visualization software to identify high-value drill sites and optimize well performance, achieving up to 20x production improvements and outperforming competitors for nearly four years through advanced analytics and cross-domain synthesis. * Developed Android applications with BLE 5.0 and cellular connectivity for telemetry systems, enabling seamless field data acquisition and remote control of industrial equipment across distributed outdoor operations. * Directed heavy equipment operations and site planning for well drilling and reclamation, leveraging mechanical expertise and electrical troubleshooting to resolve complex field failures under pressure and ensure rapid deployment of turnkey autonomous solutions. * Authored comprehensive technical documentation and quality management standards for automation workflows, accelerating team onboarding and enabling rapid learning of complex embedded systems across a fast-paced, startup-like operational environment.

Yorkshire Manor Apartments LLC & Masters Repair Group LLC | Austintown, OH Facilities & Field Operations Manager | January 2005 – December 2010 *

Orchestrated maintenance and renovation operations for 200+ residential units and light-commercial properties, implementing standardized equipment protocols and material practices that reduced maintenance costs by over \$1M annually and sustained occupancy rates above 100% through strategic lease buyouts. * Executed comprehensive HVAC, electrical, and plumbing repairs while enforcing strict safety compliance, reducing eviction rates by over 50% through enhanced tenant screening, rapid learning of building systems, and responsive facility management.

Locations Ltd | Columbus, OH Maintenance Supervisor | January 1999 – December 2005 *

Directed maintenance operations for 200 off-campus apartments, developing standardized inspection templates and turnover procedures that minimized repair needs during peak occupancy periods, improved asset longevity, and established quality management benchmarks for facility upkeep.

Technical Skills

Robotics & Automation: SCADA (Supervisory Control and Data Acquisition) Systems, Telemetry, PLC (Programmable Logic Controller) Programming, Ladder Logic, HMI, Sensor Integration, CAN-bus, PID Control, Real-Time Control, Inspection Automation, Turnkey Autonomous Solutions, Autonomous Ground Vehicles, Industry 4.0, Infrastructure Monitoring, Outdoor Deployment, Heavy Equipment Operation, Electrical Troubleshooting, Pneumatic Controls, Safety Compliance, Quality Management, Technical Documentation. Embedded & IoT: ESP32, ESP-IDF, FreeRTOS, BLE 5.0, mDNS, TLS/HTTPS, NVS, PWM, CMake, Microcontrollers, Firmware Development, Embedded Linux, Android BLE, IoT (Internet of Things) Device Management, Rapid Learning, Cross-Domain Synthesis. Data & AI: Python, PyTorch, Diffusers, LoRA Training, Multi-GPU Inference, vLLM, Prompt Engineering, SQL, SQLite, MySQL, PostgreSQL, Data Modeling, Data Mining, 3D Visualization, Real-Time Data Pipelines, API Development, Problem-Solving Under Pressure. Full-Stack & Tools: C, C++, Kotlin, Java, TypeScript, Node.js, Express, React 18, TailwindCSS, Vite, Docker, GitHub Actions, Git, VS Code Extension API, Android Studio, Jetpack Compose, Linux, Windows, Configuration Management, Collaboration, Creativity, Startup Environment Adaptability.

Key Achievements

Quadrupled oil & gas production and generated a \$23M revenue increase over two years by architecting custom SCADA/telemetry systems that automated infrastructure monitoring and optimized site selection across outdoor commercial jobsites.

Reduced production downtime by 75%, from 30–45 days to under 10 days annually, by standardizing maintenance workflows, developing benchmark testing procedures, and deploying real-time data visualization tools for predictive asset management.

Achieved up to 20x production improvements, outperforming competitors for nearly four years, through advanced process analysis, data-mining algorithms, and the integration of embedded automation controls with heavy equipment operations.

Architected a generic-core binary firmware platform for ESP32-based IoT devices, enabling a single compiled binary to deploy across 5+ hardware variants via runtime pin maps, eliminating per-variant build matrices and accelerating turnkey autonomous solution scaling.

Spearheaded cross-functional collaboration and rapid learning initiatives to deploy cutting-edge technology in dynamic field environments, consistently delivering creative problem-solving under pressure and establishing quality management standards for startup-like operational scaling.

Education

Coursework in Civil Engineering, Construction Management, HVAC/R, and Industrial Electronics with hands-on training in mechanical systems, controls, and automation | Columbus State Community College, Columbus, OH